Q.1 The matrix A^T(Transpose of A) x A is _____ if and only if the columns of A are linearly independent.

A. symmetric

B. invertible

C. scalar

D. singular

Q.2 Two vectors u and v are orthogonal to each other if _____.

A. u + v = 0

B. u . v = 0

C. u . v = 1

D. u - v = 0

Q.3 Each pair of eigenvalue and its corresponding eigenvectors provides a solution of the equation x' = Ax which is called

A. eigenfunctions

B. dynamical system

C. solution matrix

D. vector space

Q.4 If a ____ matrix has orthonormal columns, then it also has orthonormal rows.

A. rectangular

B. diagonal

C. square

D. singular

Q.5 Two vectors u and v in R^n are orthogonal if ------

A. u.v=0

B. u.v 0

Q.6 A square matrix with orthogonal columns is _____ matrix.

A. an nonivertible

B. an invertible

C. an orthonormal

D. an orthogonal

Q.7 Let s = \{ {u_1} + {u_2} + ...{u_p}\} be a basis for a subspace W of {R^n} , is also an orthogonal basis if S is an orthogonal set.

A. False

B. True

Q.8 For a real matrix A, complex Eigen values and associated Eigen vectors come in ---------

A. conjugate pairs

B. similar pairs

C. Complex pairs

D. none of the above

Q.9 If the columns of a matrix are _____ then the matrix is invertible.

A. linearly dependent

B. linearly independent

Q.10 Any orthogonal subsets of vector in an inner product space is linearly independent

A. True

B. False
